See It, Say It, Pick It

Track 2A technical report
Team: Nightcats

1. What we built

We built a local, deterministic-first manipulation planner. Apertus v1.5 8B performs language reasoning and converts an instruction into typed task slots. Open vision models inventory and segment the scene. A deterministic resolver then binds references to visible objects, checks ambiguity, physical evidence and safety obligations, and compiles the result to the official action schema.

2. Apertus and pipeline architecture

Language. Apertus produces a constrained, typed intent. A small deterministic grammar covers literal commands and safety-critical operation classes; Apertus handles the remaining semantic structure.

Perception. Qwen2.5-VL inventories objects, SAM produces masks, and Grounding DINO plus SAM performs bounded recovery for unresolved references.

Resolution and safety. A constraint-oriented resolver evaluates identity, attributes, spatial relations, uniqueness, feasibility and safety. Ambiguity becomes `ask_user`; unsupported, impossible or unsafe tasks become `refuse`.

Grounding and export. The selected first-pick mask supplies an interior `[u, v]` pixel. The schema boundary validates complete ID coverage and exports only official fields.

Local and reproducible by design

Models are staged sequentially so Apertus and the vision stack do not occupy GPU memory together. Intent, base-vision and recovery caches have independent fingerprints. All inference is local; no proprietary model or external inference API is used.

3. What was difficult for Apertus

Apertus was reliable on short literal commands but less reliable on nested relations, coordinated clauses, negation and scope. Free-form JSON also produced null-vs-empty-list and paired-field consistency errors. We addressed this with constrained generation, a flat typed schema, clause-level parsing, source-coverage checks and one bounded semantic repair. Outputs that silently drop a color, relation, quantity, exception or destination are rejected.

The harder end-to-end problem was often the language-perception contract: a correct noun such as gear can be labelled sun or `graspable_object`. The resolver combines semantic evidence with targeted recovery while preserving uncertainty.

4. Evaluation

On the public development set, the final measured automatic score was **24.0/65**: grounding ratio **0.30**, safety contribution ratio **0.45**, catch rate **0.90**, and false-alarm rate **0.50**.

The held-out run produced all **40/40** required entries: 8 executable plans, 10 clarification requests and 22 refusals. Held-out answers are private, so these are structural results rather than an accuracy claim.

5. Limitations and handoff

The main limitation is open-vocabulary perception. Generic labels, duplicate detections, weak masks and visually similar containers cause missing or ambiguous references. Complex representations remain incomplete for heterogeneous target unions, multiple simultaneous relations, scoped extrema and negative attributes. The system prefers a conservative response over an unsupported physical action.

The public repository includes pinned model revisions, Apache-2.0 licensing, official-schema dev and held-out results, a five-slide deck, Colab workflow, provenance records and commands for arbitrary scene folders. Release archives exclude credentials, model weights, evaluator answers and generated caches.

Models: swiss-ai/Apertus-v1.5-8B; Qwen2.5-VL; SAM 2.1; Grounding DINO.